



Trigonometry 1

Definition

Trigonometry is the study of triangles by applying the relations between the sides and the angles. The term "trigonometry" comes from the Greek words "trigonon" which means "triangle" and "metria" meaning "measurements".

Branches of Trigonometry

Trigonometry is divided into two branches:

Plane Trigonometry – deals with triangles in the two dimensions of the plane.

Spherical Trigonometry – concerns with triangles extracted from the surface of a sphere.

General Types of Triangles

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Right triangle – a triangle that has a right angle.

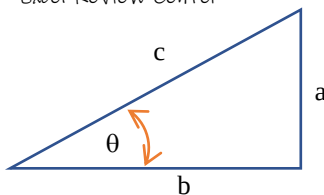
Oblique triangle – a triangle that does not have a right angle. Acute triangle and obtuse triangle are oblique triangles.

Isosceles triangle (triangle with two sides equal and two angles equal) can be a right triangle or an oblique triangle.

Solutions to Right Triangles

1. Fundamental trigonometric functions

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$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

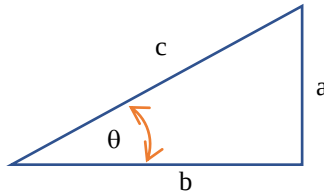
$$\cot \theta = \frac{b}{a}$$

$$\sec \theta = \frac{c}{b}$$

$$\csc \theta = \frac{c}{a}$$

2. Pythagorean theorem

"In a right triangle, the sum of the squares of the sides is equal to the square of the hypotenuse.



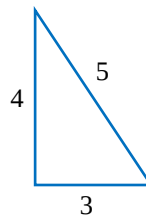
$$a^2 + b^2 = c^2$$



Pythagorean theorem is the most renowned of all mathematical theorems. It is considered as the most proved theorem in mathematics. This was formulated by Pythagoras (c.580 – c.500 B.C.) about 500 B.C.

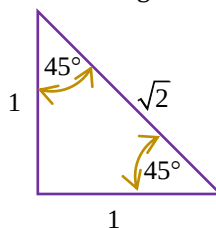
4. Special triangles

C. Egyptian triangle.

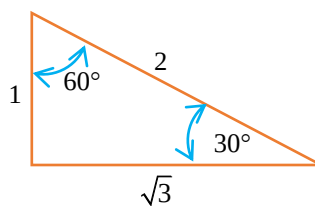


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B. Isosceles right triangle

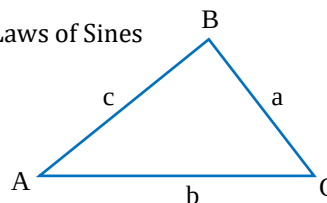


A. 30°-60° right triangle



Solutions to Oblique Triangles

1. Laws of Sines



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

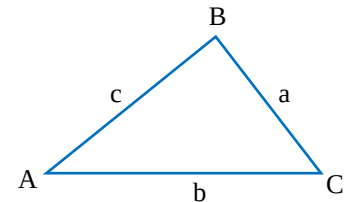
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This solution to an oblique triangle is used when the given are:

- A. two angles and one opposite side or
- B. two sides and an opposite angle

This was demonstrated by Ptolemy of Alexandria in 150 A.D.

2. Laws of Cosines



$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

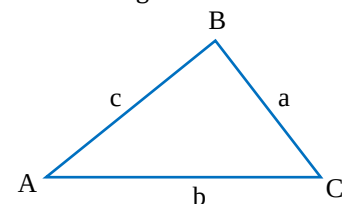
$$c^2 = a^2 + b^2 - 2ab \cos C$$

This solution to an oblique triangle is used when the given are:

- A. two sides and the included angle or
- B. three sides

This was demonstrated by the French mathematician Francois Viète (1540 – 1603).

3. Laws of Tangents



$$\frac{a-b}{a+b} = \frac{\tan \frac{1}{2}(A-B)}{\tan \frac{1}{2}(A+B)}$$

This solution to an oblique triangle is used when the given are two sides and the included angle.

This was demonstrated by the French mathematician Francois Viète (1540 – 1603). This was first described by a Danish mathematician and Physician Thomas Fincke (1561-1656) in 1583.



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